# Year 8 Mathematics
# Mini-Test 2

Baldivis
Secondary College

## Perimeter, Area and Volume

Resource Section: one A4 page of notes and calculator allowed

Name: Answers

-1 for no units

Time: 35 min    Class: ______________    / 37 marks

**Full working out must be shown to get full marks. Don't forget to add appropriate units. Attempt all the questions.**

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**Part A – Perimeter** **14 marks**

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**Question 1:** Convert the following units. [9 marks]

**a)** 120 m = 12,000 cm **b)** 4 km = 4000 m **c)** 5,450 mm = 0·00545 km

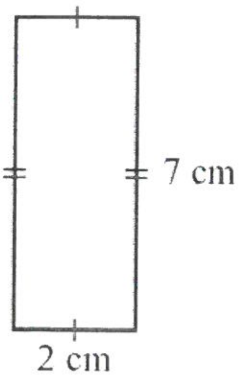
**d)** 5 $m^2$ = 0·000005 $km^2$ **e)** 12,300,000 $cm^2$ = 1230 $m^2$ **f)** 4,000 $mm^2$ = 40 $cm^2$

**g)** 687 $mm^3$ = 0·000000687 $m^3$ **h)** 46 $km^3$ = $4\cdot6\times10^{16}$ $cm^3$ **i)** 5,000,000 $cm^3$ = 5 $m^3$

**Question 2:** Find the perimeter of the following shapes. *Make sure to include the correct units.*

[1,1,1,2 = 5 marks]

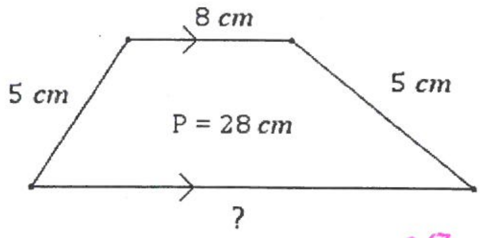
**a)**




7 + 7 + 2 + 2

= 18cm ✓

**b)**

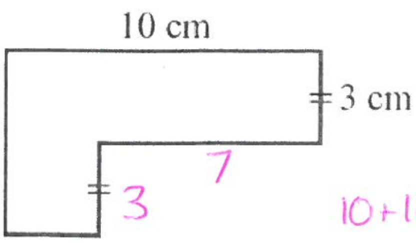



5 + 8 + 5 = 18

28 - 18

= 10cm ✓

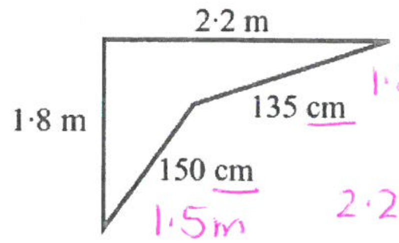
**c)**




10 + 10 + 6 + 6

= 32cm ✓

**d)**




1·35m ✓

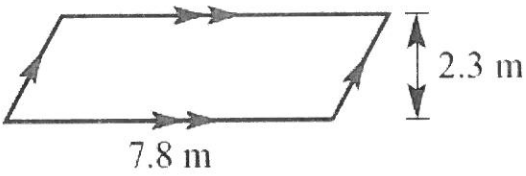
2·2 + 1·8 + 1·5 + 1·35

= 6·85m ✓

## Part B - Area **9 marks**

**Question 3:** Find the area of the following shapes. *Make sure to include the correct units.* [3 marks]

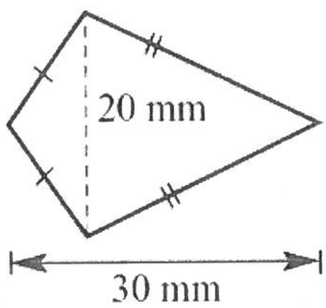
**a)**




$7.8 \times 2.3$

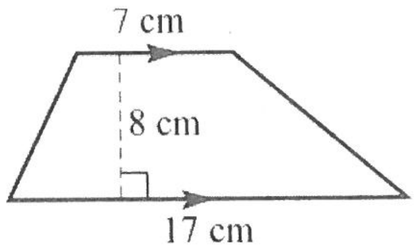
$= 17.94 m^2$ ✓

**b)**




$20 \times 30 = 600 mm^2$ ✓

**c)**



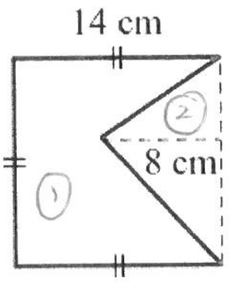

$\frac{1}{2} \times (7+12) \times 8$

$= 76 cm^2$ ✓

**Question 4:** Find the area of the composite shapes. *Make sure to include the correct units.* [6 marks]

2 mark each

**a)**




1 $14 \times 14$

$= 196 cm^2$

$\sqrt{2}$

2 $\frac{1}{2} \times 8 \times 14$

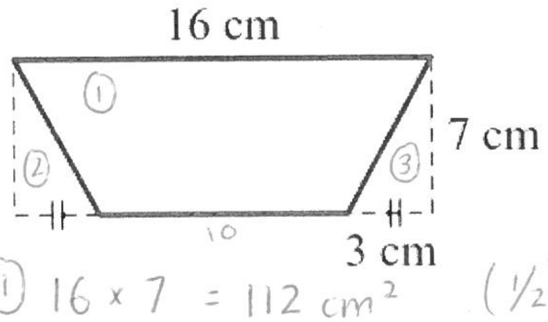
$= 56 cm^2$

$\frac{1}{2}$

$A = 196 - 56$

$= 140 cm^2$ ✓

**b)**




1 $16 \times 7 = 112 cm^2$ $(\frac{1}{2})$

2 $\frac{1}{2} \times 3 \times 7 = 10.5 \times 2 = 21$ $(\frac{1}{2})$

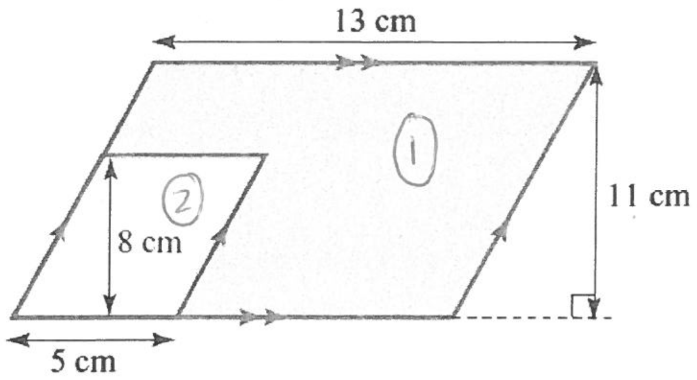
$112 - 21 = 91 cm^2$ ✓

OR

$\frac{1}{2} \times (16+10) \times 7$ ✓

$= 91 cm^2$ ✓

**c)** (Find the area of the shaded section)




① $13 \times 11 = 143cm^2$ (½)

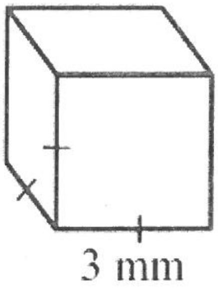
② $5 \times 8 = 40cm^2$ (½)

$143 - 40 = 103cm^2$ ✓

## Part C – Volume **8 marks**

**Question 5:** Find the volume of the following prisms. [2 marks]

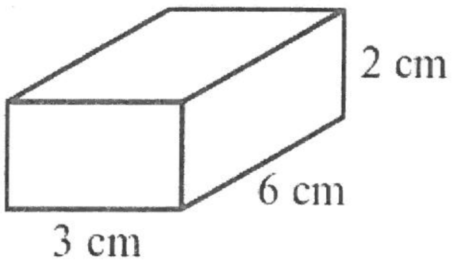
**a)**




$3 \times 3 \times 3$

$= 27mm^3$ ✓

**b)**

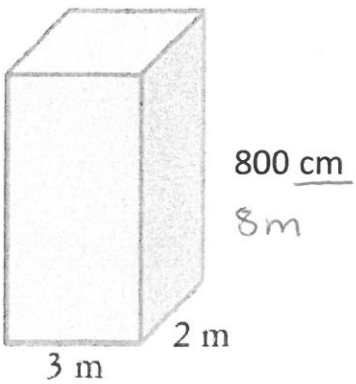



$3 \times 6 \times 2$

$= 36cm^3$ ✓

**Question 6:** Find the volume of the following prisms. *Make sure to include the correct units.* [4 marks]

**a)**

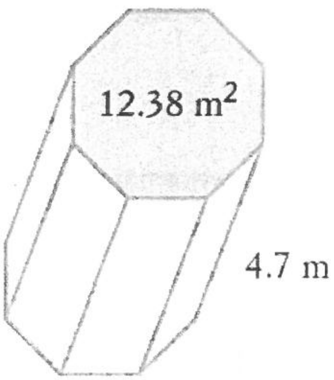



8m ✓

$8 \times 3 \times 2$

$= 48m^3$ ✓

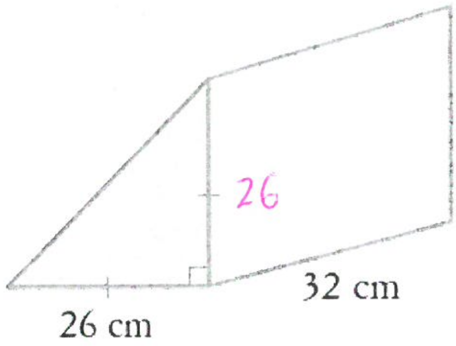
**b)**




$12 \cdot 38 \times 4 \cdot 7$ ✓

$= 58 \cdot 186m^3$

**c)**

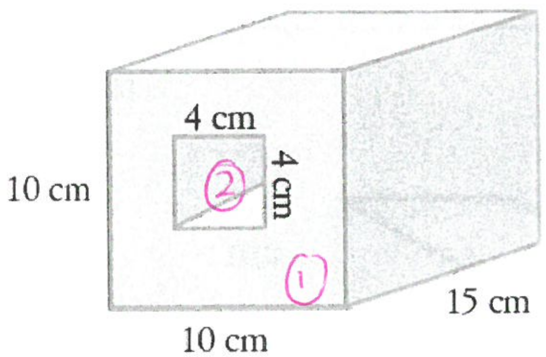



$\frac{1}{2} \times 26 \times 26 \times 32$

$= 10{,}816 cm^3$ ✓

**Question 7:** Find the volume of the following prisms. *Make sure to include the correct units.* [2 marks]

**a)**




① $10 \times 10 \times 15$
$= 1{,}500 cm^3$ $(\frac{1}{2})$

② $4 \times 4 \times 15$ $(\frac{1}{2})$
$= 240 cm^3$

$1500 - 240 = 1{,}260 cm^3$ ✓

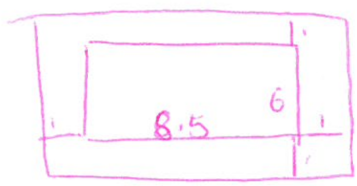
## Part D – Worded Questions 6 marks

**Question 9**: A rectangular swimming pool is 6 m × 8.5 m. [1, 1, 1, 1, 2 = 6 marks]

a) What is the area of the swimming pool?

$6 \times 8.5 = 51 m^2$ ✓

The swimming pool is surrounded by a path that is 1 m wide.




b) What are the dimensions of the swimming pool including the path?

$10.5 m \times 8 m$ ✓

c) What is the area of the swimming pool including the path?

$10.5 \times 8 = 74 m^2$ ✓

d) If the pool is 1.5 m deep, find its:

i. volume, in $m^3$

$6 \times 8.5 \times 15 = 76.5 m^2$

ii. capacity, in L

$1 m^3 = 1000 L$ ✓

$76.5 \times 1000 = 76{,}500 L$ ✓